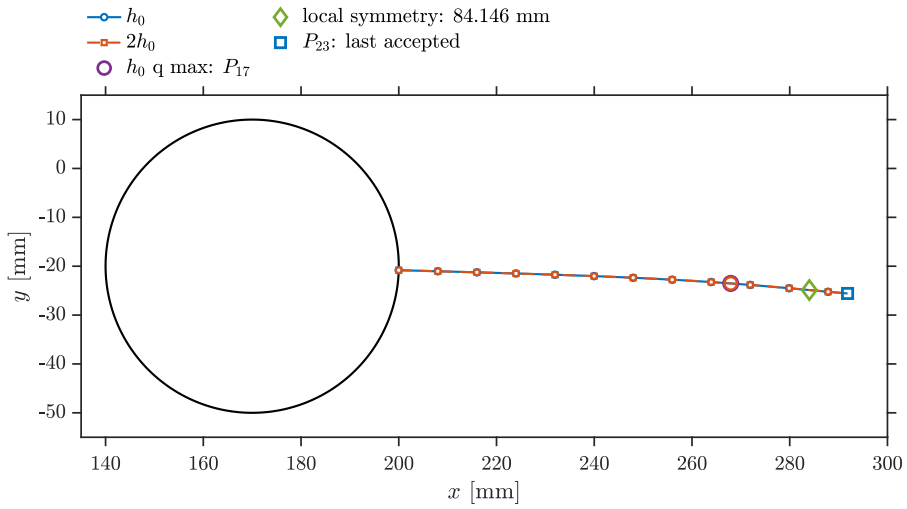
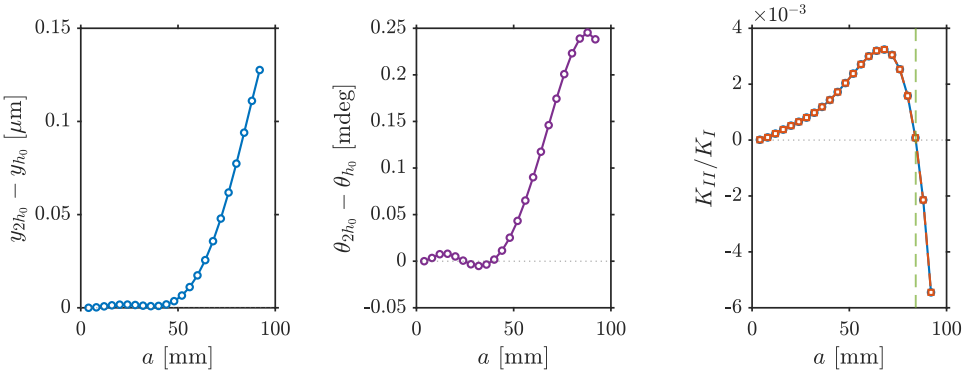


(a) Trajectories with structured-tip spacings h_0 and $2h_0$.

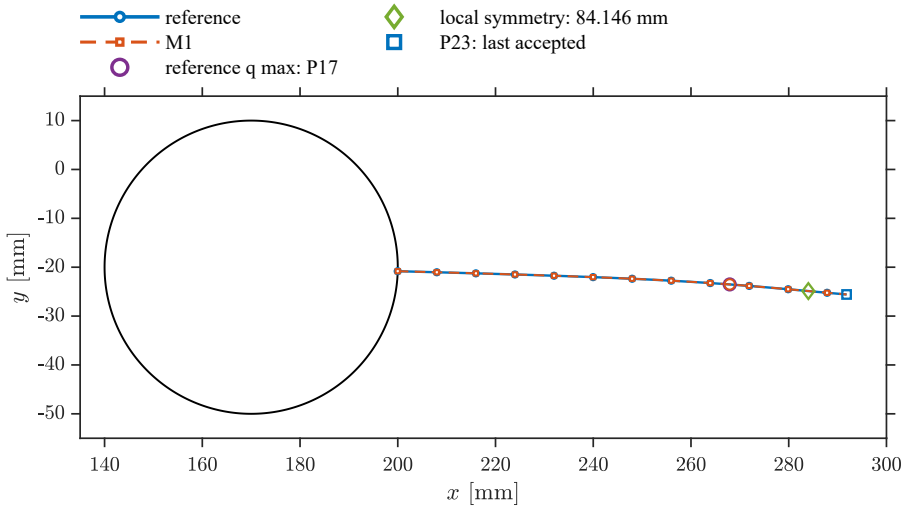


(b) Accumulated vertical tip-coordinate difference, $y_{2h_0} - y_{h_0}$.

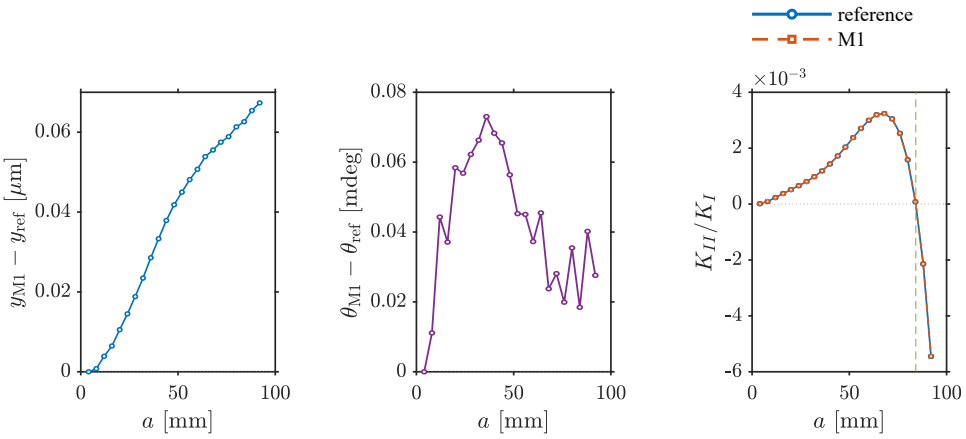
(c) Difference in the absolute crack direction in the frozen initiation frame, $\theta_{2h_0} - \theta_{h_0}$.

(d) Mode-mixity ratio $K_{\mathrm{II}}/K_{\mathrm{I}}$ along the two independently propagated trajectories.

Figure 7: Effect of structured-tip resolution on the propagated trajectory. The h_0 and $2h_0$ paths share the interaction-domain radii and prescribed exterior sizing. The diamond indicates the interpolated mode-mixity zero of the reference path.



(a) Crack trajectories with the reference and M1 exterior meshes.

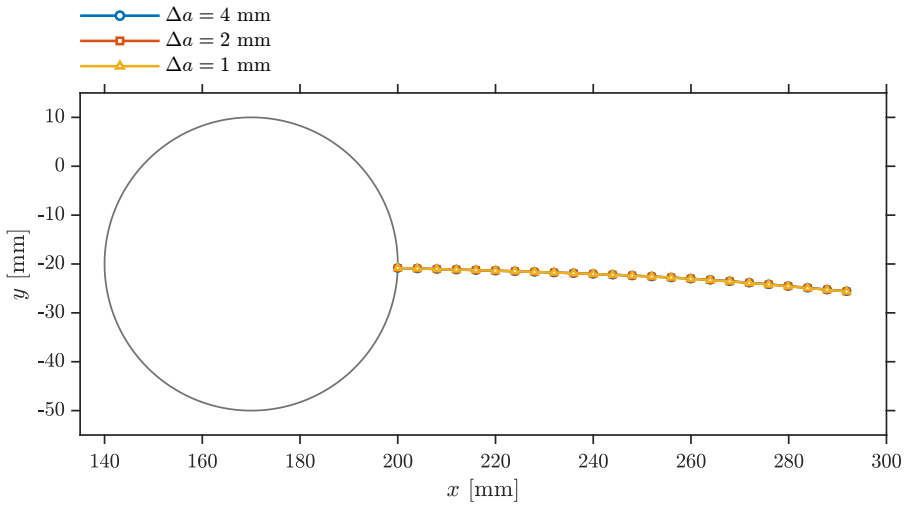


(b) Accumulated vertical tip-coordinate difference, $y_{\mathrm{M1}} - y_{\mathrm{ref}}$.

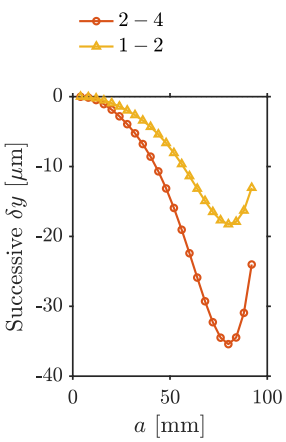
(c) Difference in the absolute crack direction in the frozen initiation frame, $\theta_{\mathrm{M1}} - \theta_{\mathrm{ref}}$.

(d) Mode-mixity ratio $K_{\mathrm{II}}/K_{\mathrm{I}}$ along the two independently propagated trajectories.

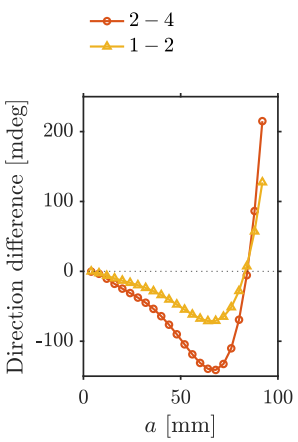
Figure 8: Effect of exterior-mesh coarsening on the crack path. The reference and M1 trajectories are propagated separately with identical near-tip discretizations and interaction-domain radii. The diamond marks the interpolated mode-mixity zero of the reference path.



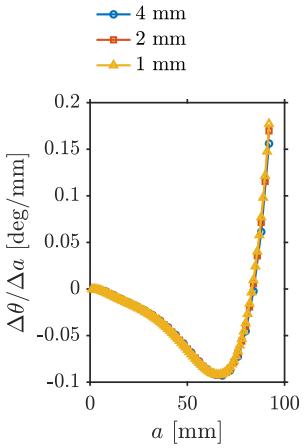
(a) Independently propagated crack trajectories for $\Delta a = 4$, 2, and 1 mm, all continued to 92 mm.



(b) Successive vertical-coordinate differences, $y_{2\,\mathrm{mm}} - y_{4\,\mathrm{mm}}$ and $y_{1\,\mathrm{mm}} - y_{2\,\mathrm{mm}}$, evaluated at exact common crack lengths.



(c) Successive differences in absolute crack direction in the frozen initiation frame.



(d) Turning density $\Delta\theta/\Delta a$ along the three independently propagated trajectories.

Figure 9: Coupled crack-increment and discretization refinement. The 4-, 2-, and 1-mm paths share dimensionless mesh and interaction-domain scales. Panels (b,c) compare paths at common crack lengths; panel (d) shows turning density on each path's native grid.